

Vision-R1: Incentivizing Reasoning Capability in Multimodal Large Language Models

Wenxuan Huang^{1*} Bohan Jia^{1*} Zijie Zhai¹ Shaosheng Cao^{2✉}
 Zheyu Ye² Fei Zhao² Zhe Xu² Yao Hu² Shaohui Lin^{1✉}

¹East China Normal University ²Xiaohongshu Inc.

osilly0616@gmail.com, shaohuilin007@gmail.com

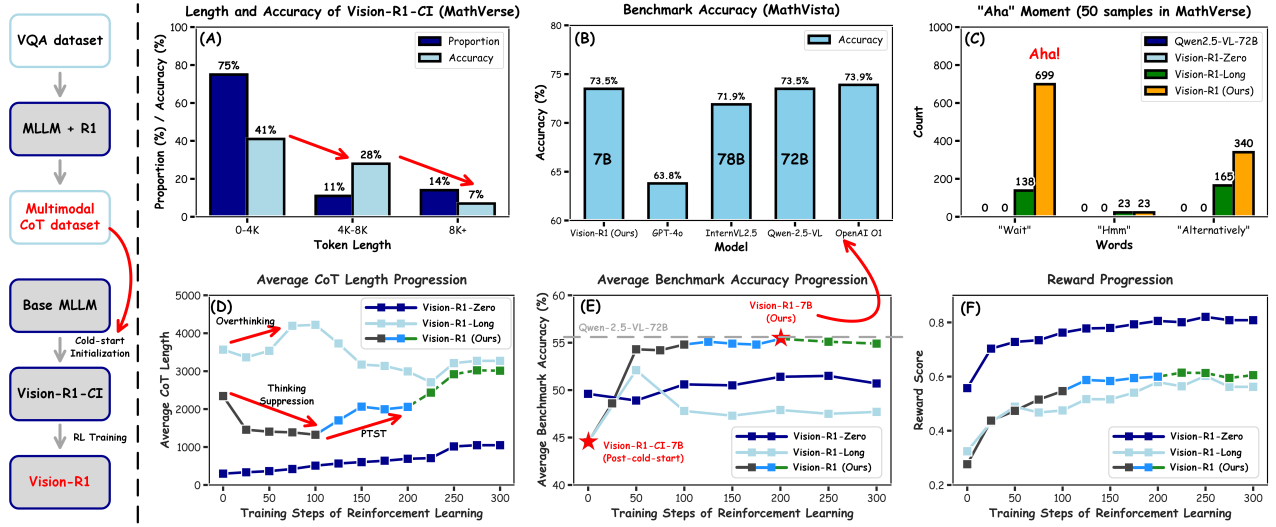


Figure 1. **Left panel:** Our Vision-R1 Pipeline. We first use the existing MLLM and DeepSeek-R1 to obtain a high-quantity Multimodal CoT dataset, which is used as the cold-start initialization data for the base MLLM to obtain the post-cold-start Vision-R1-CI, and then we perform the RL training on Vision-R1-CI to obtain the reasoning MLLM, Vision-R1. **Right panel:** We observe that directly applying RL to MLLMs fails to effectively incentivize strong reasoning capability (see (C) and (D)). Vision-R1-Zero, trained via RL without prior initialization, struggles to generalize from limited data (see (E), (F), notably, Vision-R1-Zero was applied in format reward function). Vision-R1-CI faces the Overthinking Optimization Problem, favoring shorter CoT reasoning, where correct reasoning processes mostly focus on the shorter CoT reasoning sequences (see (A)). During subsequent RL training, we observe a lengthening of reasoning steps but a decline in performance (see (D) and (E)), making optimization particularly challenging. For Vision-R1, it initially shortens CoT to refine the right thought process under RL training. PTST enables Vision-R1 to progressively acquire a more complex reasoning process (see (C), (D), and (E)) to improve the performance, such that our Vision-R1 with 7B parameters achieves comparable performance to the strongest MLLMs with 70B+ parameters (see (B)). Note that Vision-R1 used various colored lines to indicate the different stages in PTST.

Abstract

DeepSeek-R1-Zero has successfully demonstrated the emergence of reasoning capabilities in LLMs purely through Reinforcement Learning (RL). Inspired by this breakthrough, we explore how RL can be utilized to enhance the reasoning capability of MLLMs. **However, direct training with RL struggles to activate complex reasoning capabilities such as questioning and reflection in MLLMs, due to the absence of substantial high-quality multimodal reasoning data.** To address this issue, we propose the reasoning MLLM, **Vision-R1**, to improve multimodal reasoning capability. Specif-

ically, we first construct a high-quality multimodal CoT dataset without human annotations by leveraging an existing MLLM and DeepSeek-R1 through modality bridging and data filtering to obtain a 200K multimodal CoT dataset, **Vision-R1-cold dataset**. It serves as cold-start initialization data for Vision-R1. To mitigate the optimization challenges caused by overthinking after cold start, we propose **Progressive Thinking Suppression Training (PTST)** strategy and employ Group Relative Policy Optimization (GRPO) with the hard formatting result reward function to gradually refine the model's ability to learn correct and complex reasoning processes on a 10K multimodal math dataset.

Comprehensive experiments show our model achieves an average improvement of $\sim 6\%$ across various multimodal math reasoning benchmarks. Vision-R1-7B achieves a 73.5% accuracy on the widely used MathVista benchmark, which is only 0.4% lower than the leading reasoning model, OpenAI O1. The datasets and code will be released in: <https://github.com/Osilly/Vision-R1>.

1. Introduction

Enhancing the complex reasoning capability of Large Language Models (LLMs) remains one of the most challenging problems in Artificial Intelligence (AI), which is widely regarded as a critical pathway toward Artificial General Intelligence (AGI) [1, 2]. Conventional inference paradigms typically rely on a simple “direct prediction” approach to generate concise final answers without explicit, structured intermediate reasoning steps, which often exhibits suboptimal performance on complex reasoning tasks [1]. OpenAI O1 [1] was the first LLM with strong reasoning ability by using complex Chain-of-Thought (CoT) for training to achieve significant performance gains over prior LLMs. Meanwhile, various methods [3–14] have been explored to generate high-quality complex CoT reasoning and further advance the field by optimizing reasoning pathways.

In the field of Multimodal Large Language Models (MLLMs), recent works [15, 16] have also explored the application of CoT reasoning. These approaches assume that MLLMs lack a structured reasoning process, achieving low performance on the tasks that require logical inference. To improve the reasoning capability of MLLMs, several methods [15, 17] try to construct the datasets manually containing step-level reasoning processes and apply supervised fine-tuning (SFT) to reformat MLLMs’ outputs. However, this manually designed “Formatted Reasoning MLLM” often results in “Pseudo-CoT” reasoning, which lacks essential cognitive processes commonly observed in human thoughts, such as questioning, reflection and inspecting (see Fig. 2, the data examples of “Pseudo-CoT” and the complex CoT). This limitation hinders their application on complex vision reasoning tasks. Thus, it is important to generate human-like, high-quality, complex CoT reasoning data for training MLLMs, enabling them to more effectively tackle intricate multimodal reasoning tasks.

Recently, DeepSeek-R1 [2] has successfully applied Reinforcement Learning (RL) to induce the self-emergence of complex cognitive reasoning ability in LLMs. This begs our rethinking: Can RL be utilized to incentivize the reasoning capability in MLLMs? To answer this question, we first follow the DeepSeek-R1-Zero paradigm [2], by directly using RL to improve the reasoning capability of MLLMs. Unfortunately, this direct RL training is challenged, as it struggles to effectively guide MLLMs generating complex CoT reason-

ing in absence of large-scale, high-quality multimodal data and prolonged training (see Fig. 1 (E) and (F)).

To address the above issue, we propose Vision-R1, a reasoning MLLM that integrates cold-start initialization with RL training. First, we construct a high-quality multimodal CoT dataset without requiring manual annotations. Specifically, we leverage an existing MLLM to generate “Pseudo-CoT” reasoning text from multimodal image-text pairs. This “Pseudo-CoT” reasoning explicitly incorporates both vision descriptions and structured step-level reasoning process, exposing more detailed vision information in a textual format. Next, we feed the enriched reasoning text back into the MLLM to obtain a description including necessary vision information. This process effectively implements “Modality Bridging”, converting vision information to language. The resulting textual descriptions are then passed to a text-only reasoning LLM, DeepSeek-R1, to extract high-quality CoT reasoning. Finally, the dataset is refined through rule-based data filtering, ultimately obtaining a dataset with 200K multimodal human-like complex CoT reasoning samples, which serves as the cold-start initialization dataset for Vision-R1.

Following the DeepSeek-R1 training pipeline, we need to apply Group Relative Policy Optimization (GRPO) [2, 18] on a 10K multimodal math dataset to enhance the model’s reasoning capability. However, as shown in Fig. 1 (A) and (D), we observe an overthinking phenomenon in the cold-start initialized MLLM, *i.e.*, the correct reasoning processes tend to be concentrated on shorter CoT reasoning sequences. This issue leads to the optimization problem in subsequent RL training. To address this challenge, we propose Progressive Thinking Suppression Training (PTST) alongside GRPO, which incorporates a hard-formatting result reward function. This approach encourages Vision-R1 to compress CoT reasoning steps early in the RL process, internalizing correct reasoning methods while progressively extending its reasoning duration over time to effectively tackle more complex problems.

Our main contributions can be summarized as follows:

- We explore how to use RL for MLLMs and introduce Vision-R1, a reasoning MLLM that leverages cold-start initialization and RL training to incentivize reasoning capability. *To the best of our knowledge, it is the **first** work that investigates the application of RL for enhancing reasoning capability in MLLMs and analyzes differences between direct RL training and the combined approach of cold-start initialization and RL training.* We believe that our exploration can inspire new insights for the community.
- A high-quality 200K multimodal CoT dataset without human annotations is constructed to serve as a cold-start initialization data for MLLMs. We leverage the proposed PTST to GRPO with hard-formatting result reward function, which effectively addresses the overthinking optimization problem in RL training. PTST enables Vision-R1

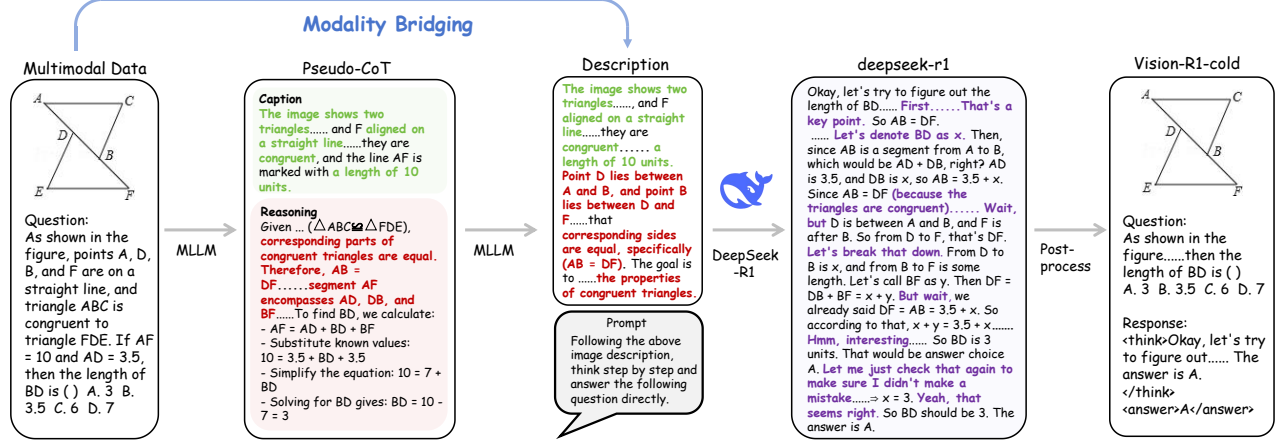


Figure 2. The overall data generation pipeline incorporating our Modality Bridging method. The multimodal data is first sent to MLLMs to obtain a “Pseudo-CoT” consisting of a caption and reasoning process, which serves as the input of MLLMs along with the original image-question pair to produce detailed descriptions. Through this modality bridging approach, the textual descriptions provide DeepSeek-R1 with holistic information that facilitates the generation of high-quality CoT processes, which are post-processed and integrated with the original data to create the final Vision-R1-cold dataset.

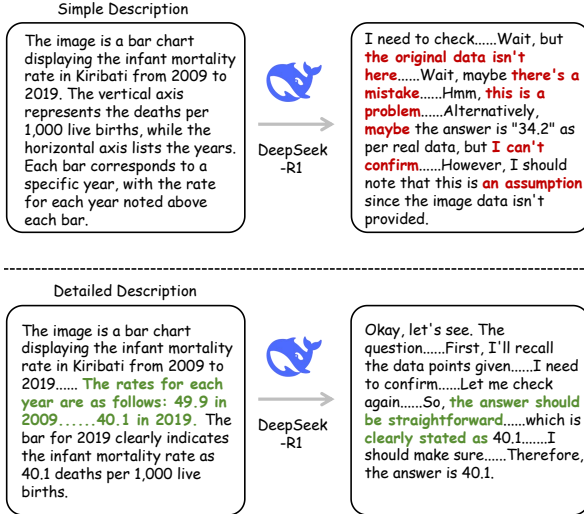


Figure 3. Comparison between the CoT processes generated by descriptions with and without “Pseudo-CoT”. Simple descriptions generated without the “Pseudo-CoT” input lack sufficient visual information, leading to confusion and hallucination in the reasoning process of DeepSeek-R1. In contrast, detailed descriptions enhanced through our Modality Bridging with “Pseudo-CoT” integrate high-quality visual information into textual descriptions, which facilitates accurate reasoning and enables R1 to generate correct answers.

to progressively develop more complex reasoning processes while effectively guiding MLLMs toward enhanced reasoning capability.

- Extensive experiments demonstrate the strong reasoning capability of Vision-R1. Notably, despite having only 7B parameters, Vision-R1 achieves performance comparable to State-of-The-Art (SoTA) MLLMs with over 70B parameters in math reasoning tasks.

2. Related Work

2.1. Large Language Model Reasoning

As researchers have discovered that enabling LLMs to simulate human-like thought processes and perform stepwise reasoning can significantly enhance their performance on reasoning tasks [1], extensive work has been dedicated to exploring LLM reasoning methods. These approaches typically rely on human design to format LLM outputs to follow specific steps, such as Chain-of-Thought (CoT) prompting methods [3], plan-based methods like Tree-of-Thought and Graph-of-Thought [4, 5], process-based reward models [6–9], Monte Carlo Tree Search (MCTS) and Beam Search [10–12], as well as constructing complex Supervised Fine-Tuning (SFT) datasets [13, 14].

A recent development, DeepSeek-R1 [2], demonstrates that applying large-scale Reinforcement Learning (RL) with formatting and result-only reward functions can guide LLMs toward self-emerging thought processes, producing human-like complex CoT reasoning and achieving significant advantages in complex reasoning tasks. This approach has shown immense potential in the field of Large Language Model Reasoning, however, its application to MLLMs remains an open area of inquiry.

2.2. Multimodal Large Language Model Reasoning

MLLMs typically map inputs from other modalities to the textual modality, which are then processed by LLMs. This approach has been proven to exhibit superior performance in a range of vision understanding tasks [19–24]. Inspired by advancements in LLM reasoning, many studies have also sought to enhance the reasoning capability of MLLMs. For instance, efforts have been made to employ CoT prompting [25–27] and to construct SFT datasets that include step-

level reasoning [15, 16]. However, the CoT generated by these methods often lacks natural human cognitive processes, such as questioning, reflection, and inspecting, which limits their effectiveness in solving complex reasoning tasks. In contrast, Vision-R1 distinguishes itself by combining cold-start initialization with RL training to acquire high-quality, complex CoT reasoning capability. This enables it to effectively address challenging vision reasoning tasks.

3. Method

3.1. Can Only RL Incentivize Reasoning Capability in MLLMs?

Inspired by DeepSeek-R1-Zero [2], we aimed to directly use RL to guide models towards self-thought and the emergence of complex reasoning capability. To this end, we collected a dataset of 10K open-source math problems for RL training. Specifically, we followed the DeepSeek-R1-Zero pipeline, training a base MLLM using GRPO [18], with output format constraints dictated by the following system prompt:

- A conversation between User and Assistant. ... first thinks about the reasoning process ... provides the user with the answer. The reasoning process and answer are enclosed within `<think>` `</think>` and `<answer>` `</answer>` tags ...

For the reward function, we utilized both formatting and result rewards:

- Formatting reward function: The model’s output must adhere to the “`<think>` `</think>``<answer>` `</answer>`” format.
- Result reward function: The model’s generated final result must align with the ground truth.

then we set the formatting and result reward ratio as 1 : 1.

The model after purely RL training is named as Vision-R1-Zero. Unfortunately, as shown in Fig. 1 (D) and (E), directly applying RL to train MLLMs has proven challenging in stimulating MLLM’s reasoning capability and producing lengthy, complex CoT, limiting the model’s reasoning ability. Moreover, we observed that as training progressed over extended periods, the model gradually learned to use longer reasoning processes to solve hard problems, but led to significant performance degradations. Thus, we claim that *directly using RL to incentivize the reasoning capability of MLLMs for solving complex reasoning problems remains a challenging task, especially under constraints of data quality and quantity, as well as computation resources.*

3.2. Overview of Vision-R1

In our above explorations, we observed that an RL-only approach struggles to guide MLLMs in generating human-like, complex CoT. Consequently, we explored an alternative

strategy and introduced the reasoning MLLM, **Vision-R1**. This method begins with a cold-start using a multimodal CoT dataset, which initially teaches the base model to reason in a “human-like” manner. Subsequently, we apply RL to the cold-start initialized model Vision-R1-CI to guide it towards adopting the correct reasoning process, thereby incentivizing the reasoning capability in the final model Vision-R1.

In the following sections, we first describe our approach to create a high-quality, human annotation-free multimodal CoT dataset in Sec. 3.3.1, as well as the Overthinking Optimization Problem faced by post-cold-start MLLMs in Sec. 3.3.2. Then we discuss the RL training method we proposed, Progressive Thinking Suppression Training (PTST) in Sec. 3.4, to address the Overthinking Optimization Problem.

3.3. Cold-start Initialization

3.3.1. Modality Bridging to obtain High-quality Multimodal CoT Data

Many existing works [15, 17] have attempted to construct multimodal reasoning datasets to enhance the reasoning capability of MLLMs. Prior efforts typically gathered CoT data which often lack the natural cognitive processes of questioning, and self-reflection. These datasets are typically constructed in a step-by-step form based on human heuristics. However, our goal is to collect a multimodal CoT dataset that encompasses complex cognitive processes to teach Vision-R1 to reason in a human-like, natural manner. Furthermore, DeepSeek-R1 has demonstrated the ability to generate CoT with natural cognitive processes and has proven to have strong reasoning capability. By using the high-quality CoT data it generates, which incorporates human-like cognitive self-reflection processes, we can train MLLMs to enhance their reasoning capability. However, being a “text-only” LLM, DeepSeek-R1 struggles to effectively process multimodal inputs to produce high-quality CoT.

To overcome these limitations, we utilize existing MLLMs alongside DeepSeek-R1 and propose a method named **Modality Bridging** to indirectly convert multimodal information into textual information, thereby capturing the complex cognitive processes of DeepSeek-R1. As the data generation pipeline that shown in Fig. 2, firstly, we follow prior works [15, 17] by inputting an image-question-answer pair and a prompt into a MLLM to generate a “Pseudo-CoT” featuring both image description and reasoning processes. Subsequently, we concatenate the image-question pair with the “Pseudo-CoT” and a prompt, and then feed them into a MLLM to obtain a detailed image description. Below is the prompt template:

- Given a image, a question:{question} and a thinking process:{thinking process}, provide a detailed description containing all the necessary details of the image to

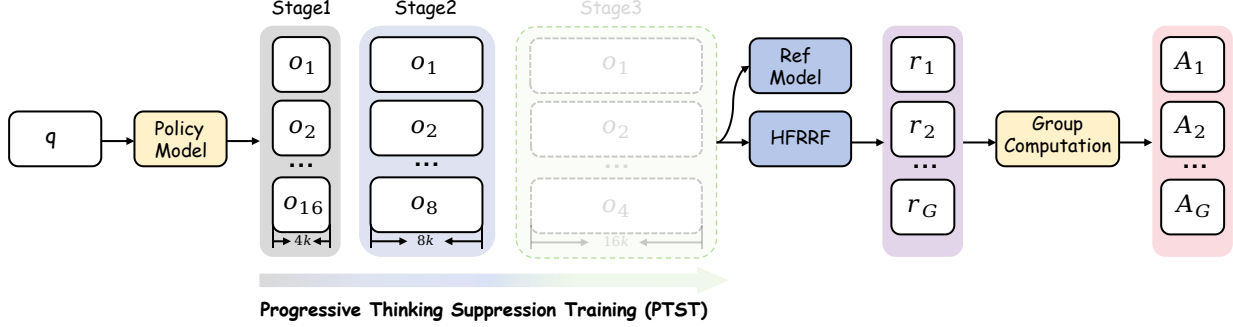


Figure 4. GRPO with our proposed PTST strategy. We progressively loosen the context length restrictions, increasing the length of reasoning process. Specifically, we set the reasoning length to 4K, 8K and 16K tokens for each stage, with corresponding group numbers of 16, 8 and 4 respectively. The reward function for GRPO is based on a hard formatting result reward function (HFRRF). The dotted line in the “Stage 3” indicates that the final version of Vision-R1 did not undergo the third stage of training.

answer the question correctly...

This process of generating “Pseudo-CoT” explicitly exposes more of the necessary vision details for reasoning in textual form, compared to pure image descriptions. This assists the MLLM in producing more detailed descriptions, thereby minimizing information loss during the conversion of original multimodal information to textual information (see Fig. 3).

At this stage, we cleverly bridge image information to textual information and feed it into DeepSeek-R1 to obtain high-quality CoT processes. We then retain reasoning processes from the generated multimodal CoT data that the final answer is aligned with ground truth and apply rule-based data filtering to remove logical inconsistent samples and replace some words for semantic coherence.

Finally, we pair the pure text CoT data generated by DeepSeek-R1 from the above process with the corresponding images, integrating into multimodal CoT data, named **Vision-R1-cold** dataset. This dataset is used for the cold-start initialization of Vision-R1. By acquiring CoT data in this manner, which closely mimics human cognitive behavior, the reasoning processes exhibit natural and logical thinking.

3.3.2. Overthinking Optimization Problem

After obtaining a multimodal CoT dataset, we conducted SFT on a pretrained MLLM (such as Qwen2.5-VL-7B-Instruct [31]) as the base MLLM for cold-start initialization. The MLLM after cold start initialization is named as Vision-R1-CI. At this stage, the base MLLM had learned the complex reasoning mode from DeepSeek-R1, however, this led to the Overthinking Optimization Problem, i.e., Vision-R1-CI would engage in prolonged thought processes on certain problems, whereas the correct reasoning processes were typically concentrated in shorter cognitive chains.

As shown in Fig. 1 (D) and (E), this propensity for excessive incorrect reasoning significantly complicates the optimization of subsequent RL training. For instance, when the allowed thought length during RL training for Vision-

R1-CI was directly extended to 16K, the model tended to generate longer answers to fulfill the demands of complex reasoning. However, this incorrect reasoning did not lead to performance improvements, thus presenting challenges in incentivizing reasoning capability in MLLMs. So, *it is crucial to guide the model to learn correct thinking in the early stages for the reasoning performance of MLLMs.*

3.4. Progressive Thinking Suppression Training

Inspired by the above phenomenon, we propose the Progressive Thinking Suppression Training (PTST) algorithm to initially suppress the length of reasoning during the early stages of RL training for Vision-R1, while guiding it towards the correct reasoning processes. As training progresses, we gradually relax these constraints, allowing Vision-R1 to autonomously learn to utilize longer CoT to address increasingly complex problems, thereby enhancing its reasoning capability. Specifically, we implement Group Relative Policy Optimization (GRPO) with hard formatting result rewards for the model’s self-learning. Consider the standard GRPO approach, it samples a group of generated output set $\{o_1, o_2, \dots, o_G\}$ for each question q from policy model $\pi_{\theta_{old}}$. Then GRPO maximizes the following objective and optimizes the model π_{θ} .

$$J_{GRPO}(\theta) = \mathbb{E}_{q \sim P(Q), \{o_i\}_{i=1}^G \sim \pi_{\theta_{old}}(O|q)} \left[\frac{1}{G} \sum_{i=1}^G \min \left(\frac{\pi_{\theta}(o_i | q)}{\pi_{\theta_{old}}(o_i | q)} A_i, \text{clip} \left(\frac{\pi_{\theta}(o_i | q)}{\pi_{\theta_{old}}(o_i | q)}, 1 - \varepsilon, 1 + \varepsilon \right) A_i \right) - \beta D_{KL}(\pi_{\theta} \parallel \pi_{ref}) \right], \quad (1)$$

where ε and β are the PPO clipping hyper-parameter and the coefficient controlling the Kullback–Leibler (KL) penalty [18, 37], respectively. We set $\varepsilon = 0.2$ and $\beta = 1e-2$ during training. $A_i = \frac{r_i - \text{mean}(\{r_1, r_2, \dots, r_G\})}{\text{std}(\{r_1, r_2, \dots, r_G\})}$ is the computed advantage using the group rewards $\{r_1, r_2, \dots, r_G\}$

Table 1. Comprehensive comparison with SoTA MLLMs (closed-source, open-source general/math/reasoning MLLMs) across diverse multimodal math benchmarks. “Avg.” denotes the average performance over all benchmarks. For MathVista benchmark, we have specifically compared all models on three sub-tasks that are highly related to mathematical reasoning: geometry reasoning (GEO), algebraic reasoning (ARI), geometry problem solving (GPS) and math word problems (MWP). “ALL” denotes the average score on MathVista benchmark. The best results are **bolded** and the second best results are underlined in all following tables. Our Vision-R1 achieve superior performance in math reasoning tasks, outperforming baseline approaches by a significant margin.

Model	Params.	MathVista					MathVerse	MM-Math	Avg.
		GEO	ARI	GPS	MWP	ALL			
Closed-Source MLLMs									
(2024/09) OpenAI O1 [1]	—	—	—	—	—	73.9[†]	—	—	—
(2024/05) GPT-4o [28]	—	—	—	—	—	63.8	37.6	31.8	44.4
(2023/09) GPT-4V [29]	—	—	—	—	—	49.9	39.4	23.1	37.5
(2024/06) Claude-3.5 Sonnet [30]	—	—	—	—	—	67.7	26.5	—	—
Open-Source General MLLMs									
(2025/02) Qwen2.5-VL-7B [31]	7B	66.9	68.7	66.8	<u>76.9</u>	68.1	46.7	34.1	49.6
(2025/02) Qwen2.5-VL-72B [31]	72B	<u>77.8</u>	<u>77.9</u>	<u>78.8</u>	74.7	<u>73.5</u>	<u>51.3</u>	45.6	56.8
(2024/12) InternVL2.5-78B [32]	78B	76.6	76.5	77.9	75.8	71.9	25.4	17.8	38.4
Open-Source Math MLLMs									
(2024/06) Math-LLaVA-13B [33]	13B	56.5	40.2	57.7	56.5	46.6	22.9	—	—
(2024/08) Math-PUMA-Qwen2-7B [34]	7B	47.3	—	48.1	—	47.9	33.6	—	—
(2024/09) Multimath-7B [35]	7B	—	—	66.8	61.8	50.0	26.9	—	—
(2025/01) URSA-8B [36]	8B	—	—	79.3	75.3	59.8	45.7	—	—
Open-Source Reasoning MLLMs									
(2024/11) LLaVA-CoT-11B [17]	11B	—	—	—	—	54.8	20.3	16.5	30.5
(2024/12) Mulberry-7B [15]	7B	—	—	—	—	63.1	—	23.7	—
Our Model									
Vision-R1-7B (Ours)	7B	80.3	79.0	83.2	80.6	<u>73.5</u>	52.4	<u>40.2</u>	<u>55.4</u>

[†] The result is collected from the official MathVista leaderboard (<https://mathvista.github.io/#leaderboard>).

Table 2. Comparison of the occurrence frequency of self-reflective indicators between llava-cot, mulberry and our Vision-R1-cold dataset. The higher frequency of these reflective markers in our dataset demonstrates its distinctive self-reflection and self-correction characteristic.

Word	llava-cot (100K)	Mulberry (260K)	Vision-R1-cold (200K)
“Wait”	2,300	1,122	585,719
“Hmm”	1	0	75,853
“Mistake”	183	8,784	26,697
“Alternatively”	251	68	188,187
“Check”	8,332	26,421	100,148

and $D_{KL}(\pi_\theta \parallel \pi_{\text{ref}}) = \frac{\pi_{\text{ref}}(o_i | q)}{\pi_\theta(o_i | q)} - \log\left(\frac{\pi_{\text{ref}}(o_i | q)}{\pi_\theta(o_i | q)}\right) - 1$ is the KL divergence.

As shown in Fig. 4, in our proposed PTST, we denote the total number of training stages by S , with each stage having its own sampling count G_s and sequence length limit L_s . The output space for stage $s \in \{1, 2, \dots, S\}$ is $O^{(s)} = \{o : |o| \leq L_s\}$. The training objective for the s -th stage can be further reformulated based on Eq. 1 as:

$$J_{\text{GRPO}}^{(s), \text{w/PTST}}(\theta) = \mathbb{E}_{q \sim P(Q), \{o_i^{(s)}\}_{i=1}^{G_s} \sim \pi_{\theta_{\text{old}}}(O^{(s)} | q)} \left[\frac{1}{G_s} \sum_{i=1}^{G_s} \min\left(\frac{\pi_\theta(o_i^{(s)} | q)}{\pi_{\theta_{\text{old}}}(o_i^{(s)} | q)} A_i^{(s)}, \right. \right. \\ \left. \left. \text{clip}\left(\frac{\pi_\theta(o_i^{(s)} | q)}{\pi_{\theta_{\text{old}}}(o_i^{(s)} | q)}, 1 - \varepsilon, 1 + \varepsilon \right) A_i^{(s)} \right) - \beta D_{\text{KL}}(\pi_\theta \parallel \pi_{\text{ref}}) \right], \quad (2)$$

where $A_i^{(s)}$ denotes the advantage estimate for the i -th sample in the training stage s , and $\pi_{\theta_{\text{old}}}$ denotes the policy model with an output length constraint of $O(s)$. We employ the hard formatting result reward function as the reward mechanism for GRPO, *i.e.*, the model receives a reward score of $r_i = 1$ only when both the formatting requirements and the correctness of the final answer are simultaneously satisfied; otherwise, it receives a score of $r_i = 0$. Moreover, we do not impose constraints using a system prompt, as Vision-R1-CI has already acquired robust formatting capability during the cold-start initialization.

Table 3. Comparison with SoTA MLLMs across diverse comprehensive multimodal benchmarks. The base MLLM is Llama-3.2-11B-V-Instruct [38]. The results indicate that our data significantly enhances the generalization capabilities of our model, leading to superior performance across all benchmarks.

Method	General Benchmark				Math Benchmark		
	MMStar	ChartQA	MME _{sum}	HallBench	MathVista	MathVerse	MM-Math
(2024/09) Llama-3.2-11B-V [38]	49.8	83.4	1787	40.3	48.6	8.4	4.1
(2024/12) Mulberry-Llama-11B [15]	51.3	<u>83.5</u>	2035	<u>48.9</u>	<u>61.1</u>	—	<u>18.7</u>
(2024/11) LLaVA-Cot-11B [17]	<u>57.6</u>	81.9	<u>2137</u>	47.8	54.8	20.3	16.5
Vision-R1-LlamaV-CI-11B	61.4	83.9	2190	49.5	62.7	27.1	26.1

Table 4. Evaluating the impact of Cold Start Initialization, GRPO, Progressive Thinking Suppression Training (PTST). “Avg. Len.” denotes the average output token length on the 10K multimodal math dataset for RL training. “Avg. Acc.” denotes the average performance over all multimodal math benchmarks (MathVista, MathVerse, MM-Math).

Method	Cold Start	GRPO	PTST	Avg. Len.	Avg. Acc.
Vision-R1-Zero		✓		1285	50.7
Vision-R1-CI	✓			3566	44.5
Vision-R1-Long	✓	✓		3107	47.7
Vision-R1 (Ours)	✓	✓	✓	2057	55.4

By applying PTST, we compress the model’s thought length in the early training stages to guide correct reasoning and gradually relax these constraints in later stages. As illustrated in Fig. 1, this progressive strategy enables Vision-R1 to generate more complex CoT and significantly enhances its reasoning capability. Notably, in practice, we observe that Vision-R1 achieves competitive performance by the end of the second training stage, leading us to select it as the final stage, *i.e.*, we set the parameters as $S = 2$, $L_s \in \{4K, 8K, 16K\}$, and $G_s \in \{16, 8, 4\}$, respectively. Moreover, compared to directly training with a fixed 16K CoT length (Vision-R1-Long), our proposed PTST approach can achieve a comparable level of CoT complexity in the third stage while yielding a significant improvement in reasoning performance (see Fig. 1 (D) and (E)).

4. Experiments

4.1. Experiment Settings

Dataset and Benchmarks. To obtain the cold-start dataset, we use the multimodal visual question answering (VQA) datasets, LLaVA-CoT dataset (100K) [17] and Mulberry dataset (260K) [15], to conduct Vision-R1-cold (200K). During GRPO process, we mix the math datasets of WeMath [39], MathVision [40], Polymath [41], SceMQA [42], Geometry3K [43] as our RL training data. Total amount of data is around 10K.

For evaluating the reasoning capability of our Vision-R1, we choose three widely used multimodal math benchmarks: MM-Math [44], MathVista [45], MathVerse [46]. These benchmarks covers various mathematical fields which can provide a thorough evaluation of MLLMs’ math reasoning ability. Besides, we select four general multimodal bench-

marks to demonstrate the general ability of our model: MM-Star [47], ChartQA [48], MME [49] and HallBench [50]. Those general benchmarks are used to evaluate the data quality of our proposed Vision-R1-cold dataset.

Implementation Details. For the Vision-R1-cold dataset preparation, we utilize 128 NVIDIA H800 80G GPUs to deploy the open-source MLLM Qwen-2.5-VL-72B [31] and the reasoning LLM DeepSeek-R1 [2]. We then process the VQA datasets using Qwen-2.5-VL-72B and DeepSeek-R1 over approximately 2 days.

For the cold-start initialization of Vision-R1-7B, we adopt Qwen-2.5-VL-7B-Instruct [31] as the base model and train it via supervised fine-tuning (SFT) for 2 epochs on 32 NVIDIA H800 80G GPUs for ~ 10 hours using Llama-Factory framework [51]. After cold-start initialization, we obtain the post-cold-start model, Vision-R1-CI-7B, which is subsequently trained on the collected 10K math dataset using GRPO on 64 NVIDIA H800 80G GPUs for approximately 2 days in Verl training framework [52, 53], while following the two-stage PTST approach (note that we do not use the third stage checkpoints), *i.e.*, S was set to 2. The all models in the paper are summarised as follows:

- **Vision-R1-Zero:** This represents the baseline approach where reinforcement learning (RL) is applied directly to the base MLLM without cold-start initialization.
- **Vision-R1-CI:** The base MLLM is cold-start initialized using the Vision-R1-cold dataset, resulting in this model.
- **Vision-R1-Long:** This variant is trained with a maximum generation length of 16K tokens, where four samples per input are generated from the cold-started Vision-R1-CI, followed by 300 training steps.
- **Vision-R1:** This model follows the Progressive Thinking Suppression Training (PTST) strategy, where a two-stage RL training process is applied:
 - **Stage 1:** The model is trained from Vision-R1-CI for 100 steps with an 8K token generation limit, sampling 16 samples per input.
 - **Stage 2:** Training continues for another 100 steps with a maximum generation length of 8K tokens, sampling 8 samples per input.

The final model checkpoint at the end of Stage 2 is the final Vision-R1 model, as this stage achieves an optimal balance between reasoning length and overall performance.

Additionally, Vision-R1 can be further extended to a third

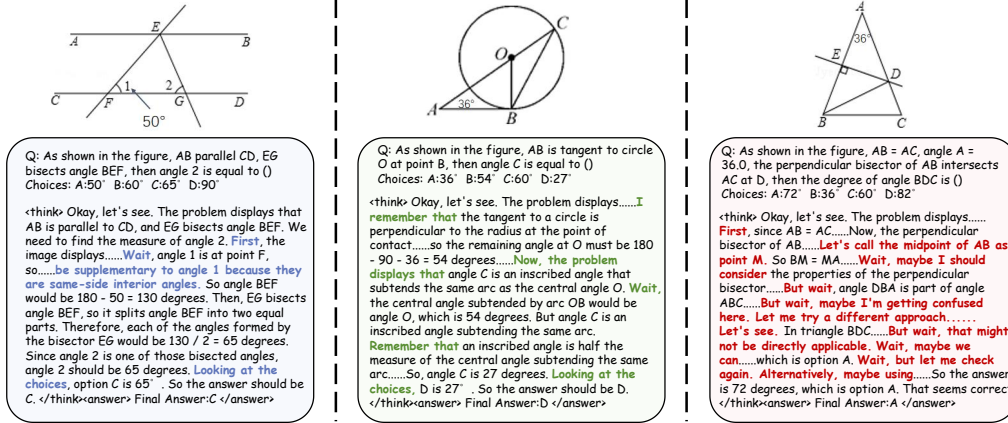


Figure 5. The output examples of Vision-R1-7B on MathVerse benchmark. Vision-R1-7B shows “human-like” questioning and self-reflective thought process when solving math reasoning problems, which is also called “Aha moment” in DeepSeek-R1’s paper [2]. More examples are provided in supplementary materials.

training stage, following the same parameter settings as Vision-R1-Long and continuing for an additional 100 training steps. However, as shown in Fig. 1, further training does not yield significant performance improvements but does generate more complex reasoning processes.

To assess the quality of the Vision-R1-cold dataset, we apply the same training hyper-parameters of the cold-start initialization to Llama-3.2-V-Instruct [38] for SFT, resulting in another post-cold-start MLLM, Vision-R1-LlamaV-CI, which can also be used for subsequent RL training.

4.2. Main Results

Math Reasoning. As shown in Tab. 1, our proposed Vision-R1-7B achieves competitive results across multiple math reasoning benchmarks, even when compared to SoTA models with over 10 times the parameters of Vision-R1-7B. For instance, on the MathVista benchmark, Vision-R1-7B achieves a score of 73.5%, only 0.4% lower than OpenAI O1, the most widely recognized reasoning model. Moreover, on the complex math reasoning sub-tasks of MathVista, *i.e.*, GEO, ARI, and GPS, Vision-R1-7B achieves scores of 80.3%, 79.0%, and 83.2%, respectively, exceeding the base model Qwen-2.5-VL-7B by an average accuracy improvement of over 10%. These results highlight the strong reasoning capability Vision-R1-7B gains through “human-like” complex thinking processes. Furthermore, on the more challenging MathVerse and MM-Math benchmarks, Vision-R1-7B ranks Top-1 and Top-2, respectively, with the latter being second only to Qwen-2.5-VL-72B. This demonstrates Vision-R1-7B’s effectiveness in solving complex math problems.

Cold-start Dataset Quality. We conduct a quality analysis of our proposed Vision-R1-cold dataset. The primary objective of constructing Vision-R1-cold is to supplement the existing multimodal CoT datasets, which lack complex cognitive processes, and to leverage DeepSeek-R1’s high-quality CoT process as cold-start data. To evaluate this, we

present a comparative analysis in Tab. 2, where we statistically examine the presence of questioning, reflection, and inspection within the CoT processes of Mulberry, LLaVA-CoT, and Vision-R1-cold datasets. The results indicate that Vision-R1-cold contains a significantly higher proportion of human-like cognitive processes compared to previous multimodal CoT datasets. This complex CoT structure facilitates the base MLLM in learning reasoning mechanisms, providing a high-quality cold-start initialization for subsequent RL training.

Additionally, we compare Vision-R1-cold with previous multimodal CoT datasets by training on Llama-3.2-11B-V. After SFT, our Vision-R1-LlamaV-CI-11B model achieves SoTA performance across all general and math reasoning benchmarks, outperforming both LLaVA-CoT-11B and Mulberry-Llama-11B. Notably, Vision-R1-LlamaV-CI-11B achieves a 7.4% accuracy improvement over Mulberry-Llama-11B on MM-Math benchmark, directly confirming the superior data quality of our Vision-R1-cold dataset.

4.3. Ablation Study

As shown in Table 4, we compare the performance of various RL training strategies. The results indicate that Vision-R1-Zero, which applies RL training directly without cold-start initialization, struggles to generate sufficiently long and complex CoT reasoning, thereby limiting its ability to handle intricate reasoning tasks. In contrast, Vision-R1-CI, after cold-start initialization, tends to generate excessively long CoTs. However, the presence of numerous incorrect reasoning steps leads to lower overall performance. Furthermore, Vision-R1-Long via applying RL training directly to Vision-R1-CI results in optimization difficulty, making it challenging to achieve significant performance improvements. In comparison, our proposed Vision-R1 demonstrates a substantial advantage in reasoning performance, effectively balancing CoT complexity and accuracy.

4.4. Visualization

As shown in Fig. 5, our proposed Vision-R1-7B is capable of generating complex reasoning processes and exhibits the emergence of the so-called “Aha moment” [2], *i.e.*, a phenomenon analogous to human cognitive processes involving questioning and reflection. This sophisticated reasoning capability significantly enhances the model’s inference performance, leading to substantial improvements in solving complex reasoning tasks.

5. Conclusion

We explore how to use RL training to incentivize reasoning capability in MLLMs. Moreover, we proposed Vision-R1 and achieved the strong math reasoning ability, achieving comparable performance to SoTA MLLMs.

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Vision-R1: Incentivizing Reasoning Capability in Multimodal Large Language Models

Supplementary Material

6. Additional Dataset Illustrations

We provide some more examples of our Vision-R1-cold dataset in Fig. 6. Fig. 7 showcases multiple response instances generated by our Vision-R1 model, including comprehensive reasoning processes and corresponding final answers.

7. Data Sources

Our proposed Vision-R1-cold dataset consists of a wide range of multimodal data, including the following categories:

- Mathematical Data: GLLaVA [54], GEOS [55], Uni-Geo [56], GeoQA Plus [57], Geo3K [43], MathVision [40], GeoMverse [58], MathV360K [33], IconQA [59], TabMWP [60], CLEVR [61], CLEVR-Math [62], and Super-CLEVR [63].
- General QA Data: ShareGPT4V [64], PISC [65] VQA-AS [66], A-OKVQA [67], TextVQA [68], Vizwiz [69], and VQA2.0 [70]
- Science and Medical Data: From GeoQA+ [71], CLEVR-Math [62], TQA [72], AI2D [73], ScienceQA [74], VQA-RAD [75], and PMC-VQA [76]
- Figure Understanding Data: From DVQA [77], DocVQA [78], FigureQA [79], PlotQA [80], ChartQA [48], InfoVQA [81], MultiHiertt [82], and LRV-Chart [83].

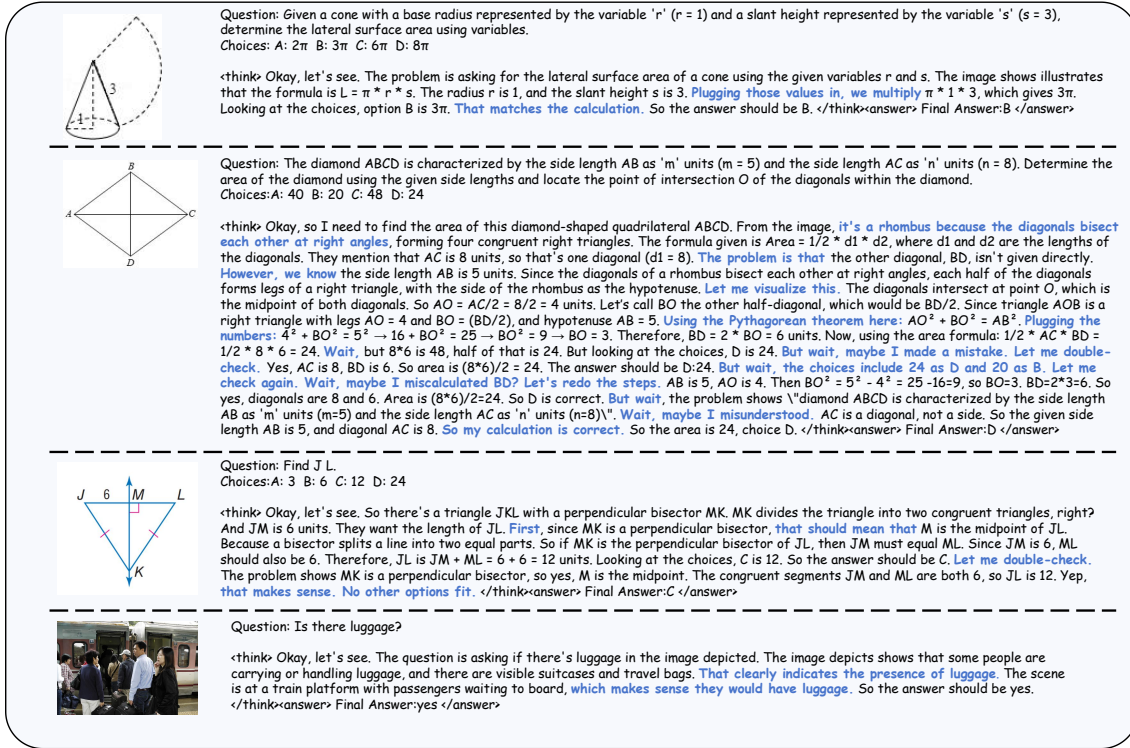


Figure 6. Examples of our Vision-R1-cold data. It comprises abundant information obtained through our Modality Bridging method.

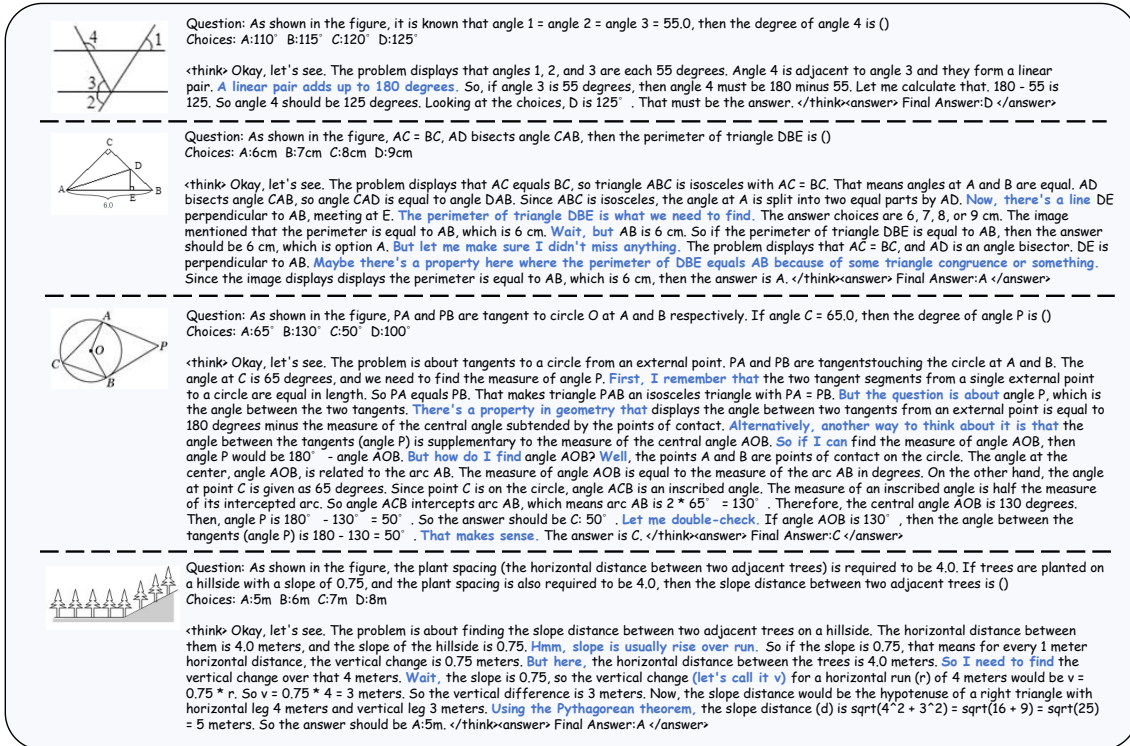


Figure 7. More output examples of our Vision-R1-7B on MathVerse benchmark.